

UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Artificial Intelligence

TOPIC 4: SUPERVISED LEARNING

SAIA 2113 – Machine Learning
Session 2 2024/2025

Prepared by:
Dr. Adam Bin Mohd Khairuddin
adam.mk@utm.my

Innovating Solutions

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

Introduction - What is supervised learning?

Supervised learning is the most common type of machine learning.

Three strategy to implement supervised learning:-

1. Collect a dataset of labeled training examples.
2. Teach the model to output accurate predictions on the dataset.
3. When the model sees new, similar data, it will also be accurate.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Introduction - What is supervised learning?

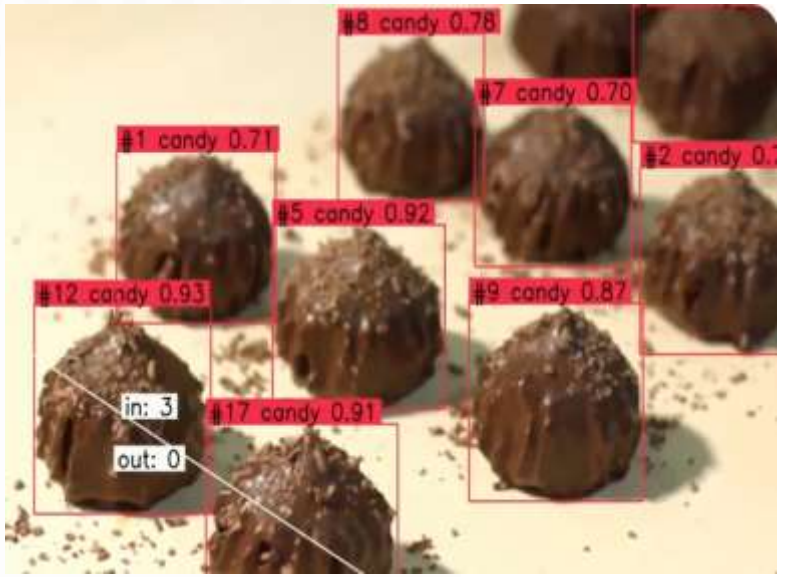
Labeled training examples:



Detection



Tracking



Counting



Analysis

Source: https://roboflow.com/

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

Components of a Supervised Machine Learning

- There are three key components of supervised machine learning: a dataset, a learning algorithm and a predictive model.
- A dataset and a learning algorithm is defined to apply supervised learning:

Dataset + **Learning Algorithm** → **Predictive Model**

- The result of running the learning algorithm on the dataset is a predictive model.
- The predictive models maps inputs to the targets.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

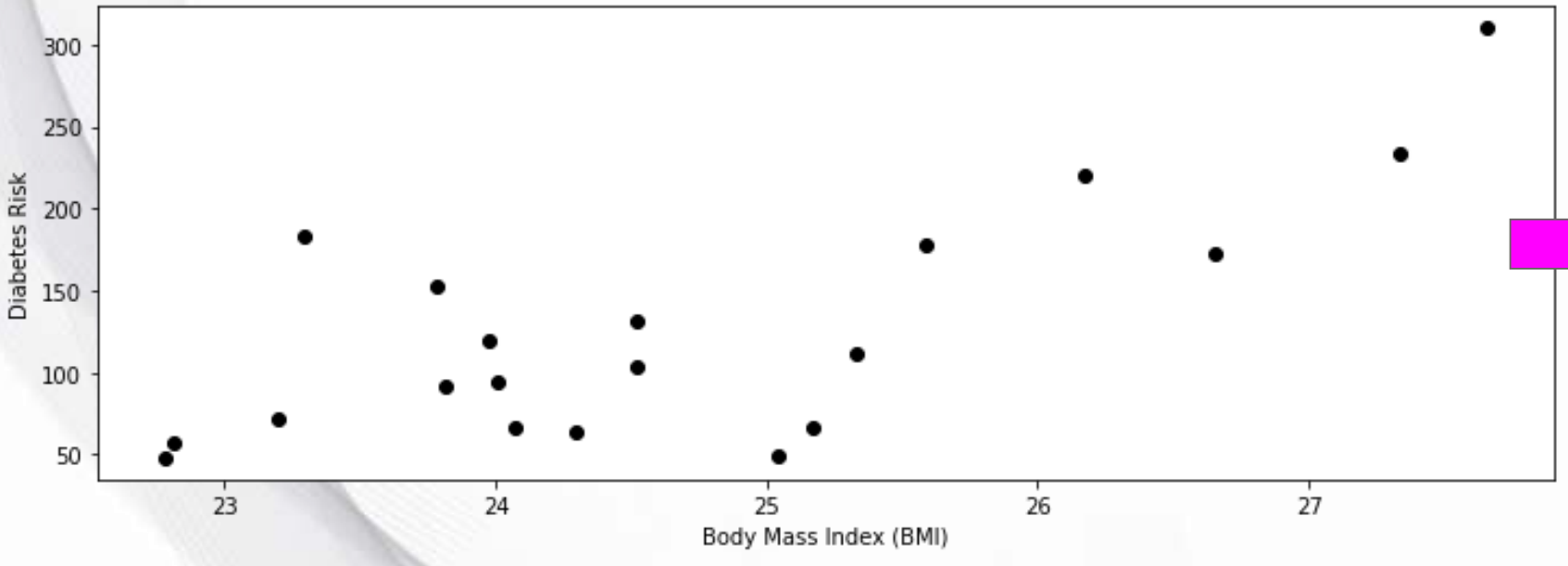
Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

A Supervised Learning Dataset

- For example, let us use the simple dataset of medical patients: (https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_diabetes.html)
- This dataset contains the BMI of medical patients and an estimate of diabetes risk (from 0 - 400).
- We want to investigate how BMI affects an individual's diabetes risk.



Diabetes risk grows as the patient's BMI increases.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

A Supervised Learning Algorithm

A supervised learning algorithm as having two components: a model class and an optimizer.

The model class

- The model represents a set of possible relationships between BMI and risk that we believe to be true.
- Assume that risk is a linear function of BMI:

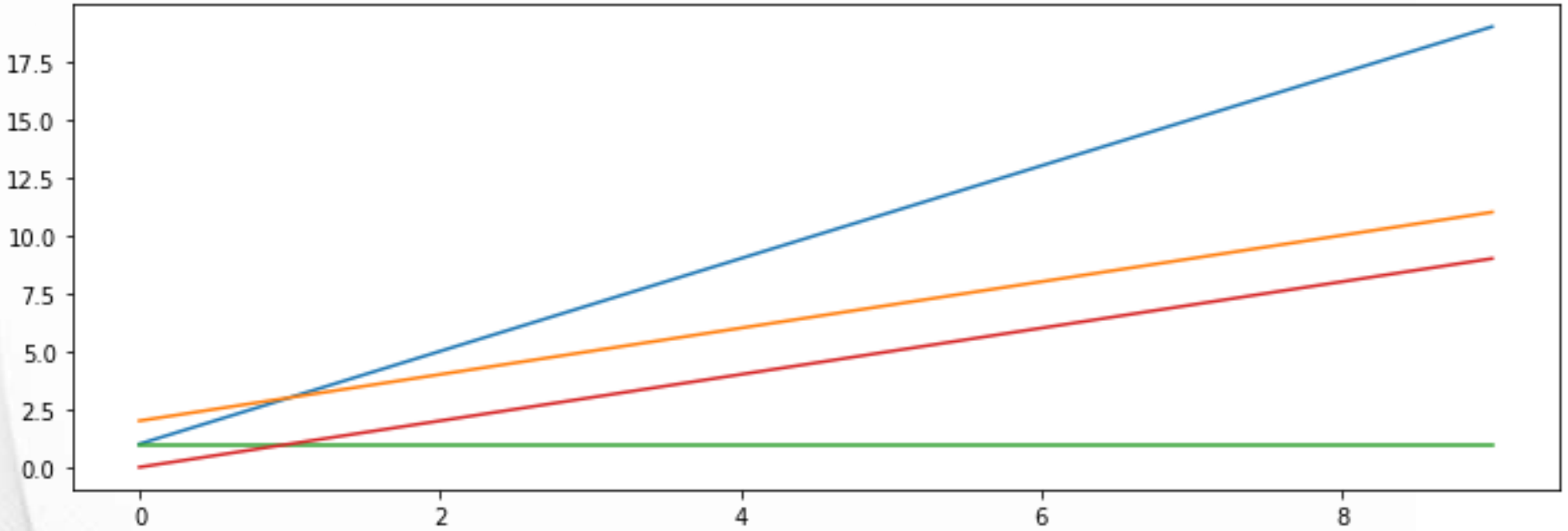
$$y = \theta_1 . x + \theta_0 ,$$

where:

x - BMI (independent variable)
y - Diabetes risk score (dependent variable)
θ_1, θ_0 - slope and the intercept of the line relates x to y.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

A Supervised Learning Algorithm



The supervised learning algorithm will choose the linear relationship fits the training data well.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

A Supervised Learning Algorithm

The optimizer

- The optimizer performs the search procedure.
- The search procedure means that we want to maximize the fit between the model and the data over the space of all possible models.
- In the case of the diabetes dataset, sklearn.linear_model library is used to find the best slope and intercept that fits the data well.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

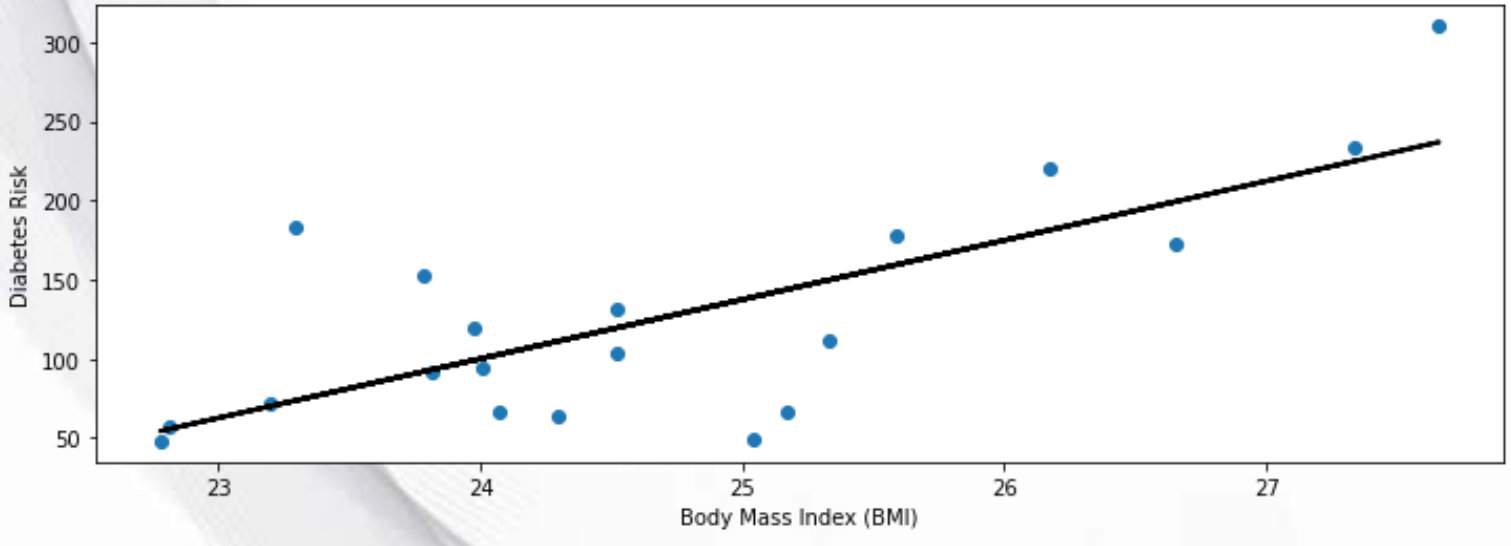
A Supervised Learning Model

- The predictive model, f is defined as:

$$f(x) = \theta^*_1 . x + \theta^*_0,$$

where:

x - BMI (independent variable)
y - Diabetes risk score (dependent variable)



The linear model is close to the observed data and captures the trend.

*Higher BMIs are associated with higher diabetes risk.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

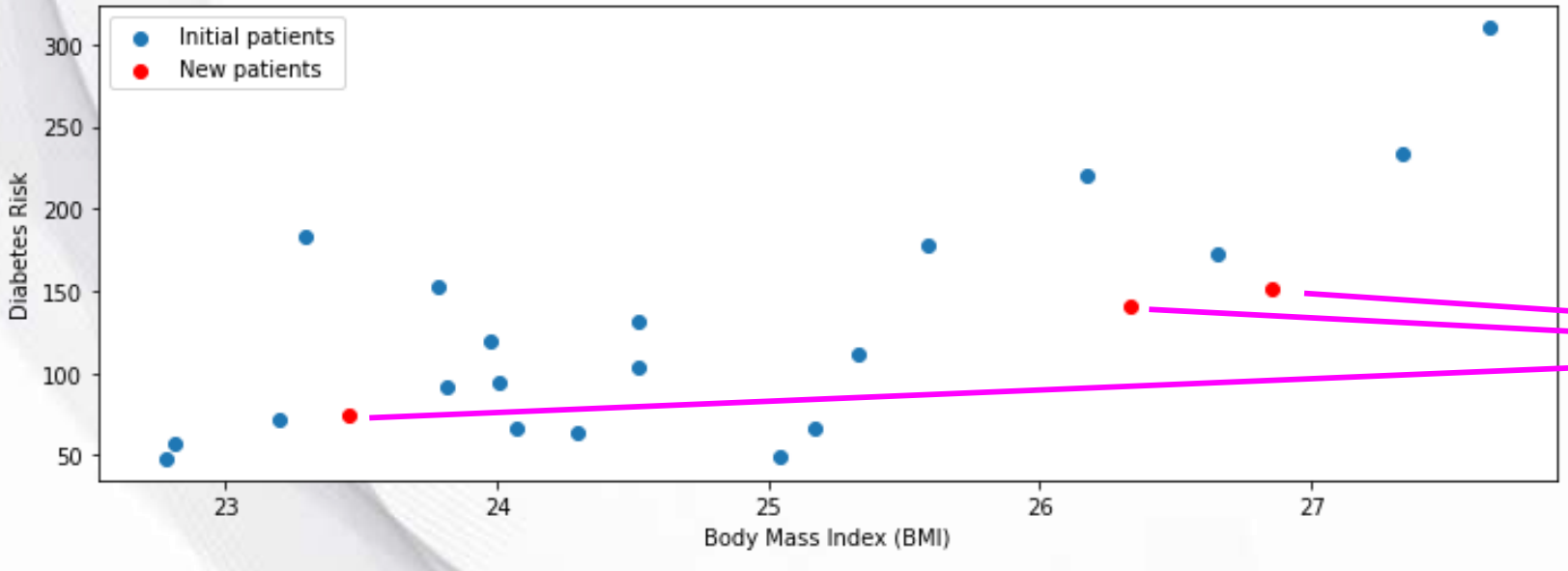
www.utm.my

Making New Predictions

- Given a new dataset of patients with known BMI, the predictive model can be used to estimate their risk.

- Formally, given an x_{new}, we can output prediction y_{new} as:

$$y_{new} = f(x_{new}) = \theta^*_1 \cdot x_{new} + \theta^*_0,$$

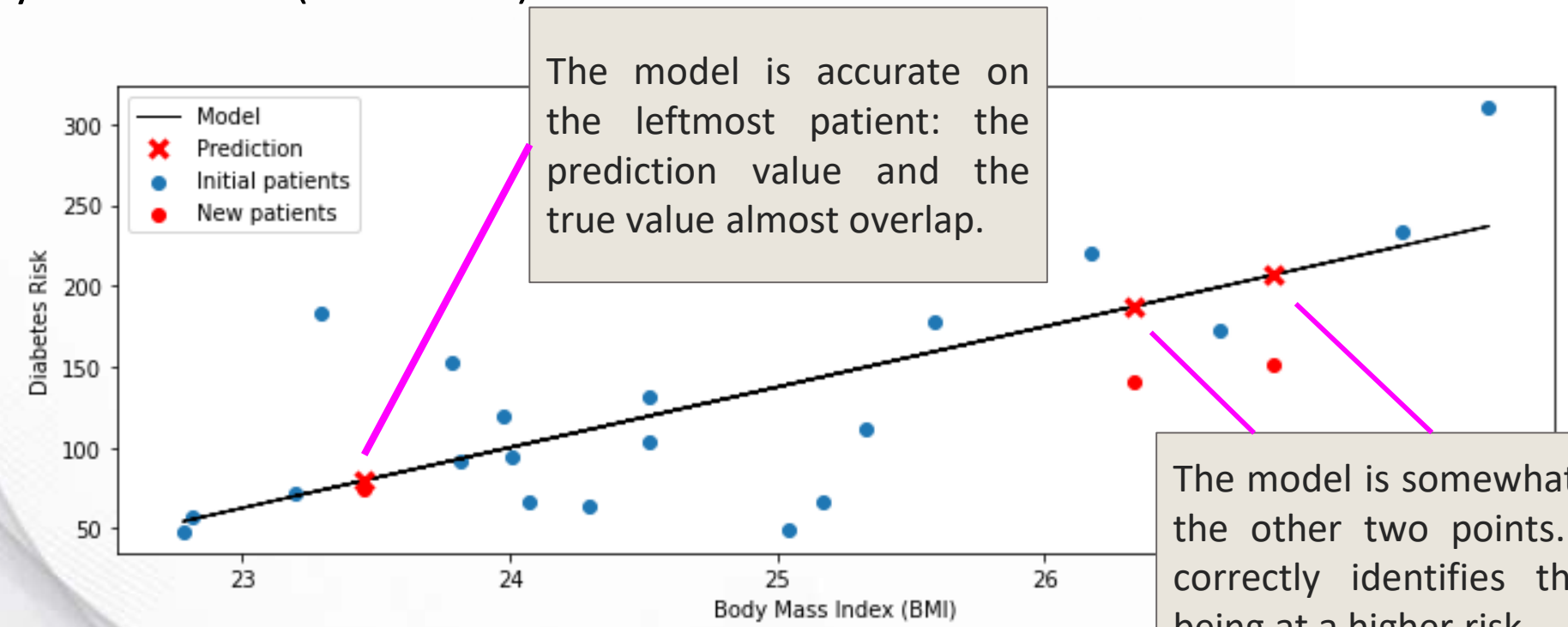


Three new patients data that we haven't seen before are added.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Making New Predictions

- The linear model provides an estimate of the diabetes risk for these patients.
- $f(x)$ is denoted by a red cross. (predicted value)
- y is denoted by a red circle. (true value)



The model is accurate on the leftmost patient: the prediction value and the true value almost overlap.

The model is somewhat off on the other two points. It still correctly identifies them as being at a higher risk.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

Why Supervised Learning?

Supervised learning is a tool that is applicable on tasks such as:

- Predictive modeling. Supervised learning can outperform human experts when more data are gathered to characterize a patient. For instance, age, gender, historical blood pressure, medical notes.

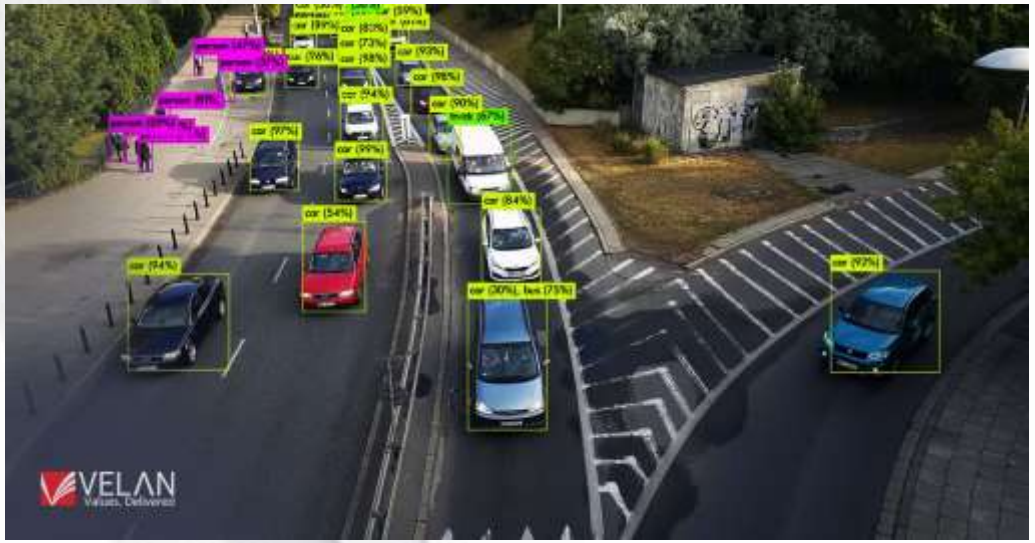
- Understanding the mechanisms through which input variables can affect targets. For example, we inspected the slope of the model and inferred from the data that a high BMI tends to increase diabetes risk.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Why Supervised Learning?

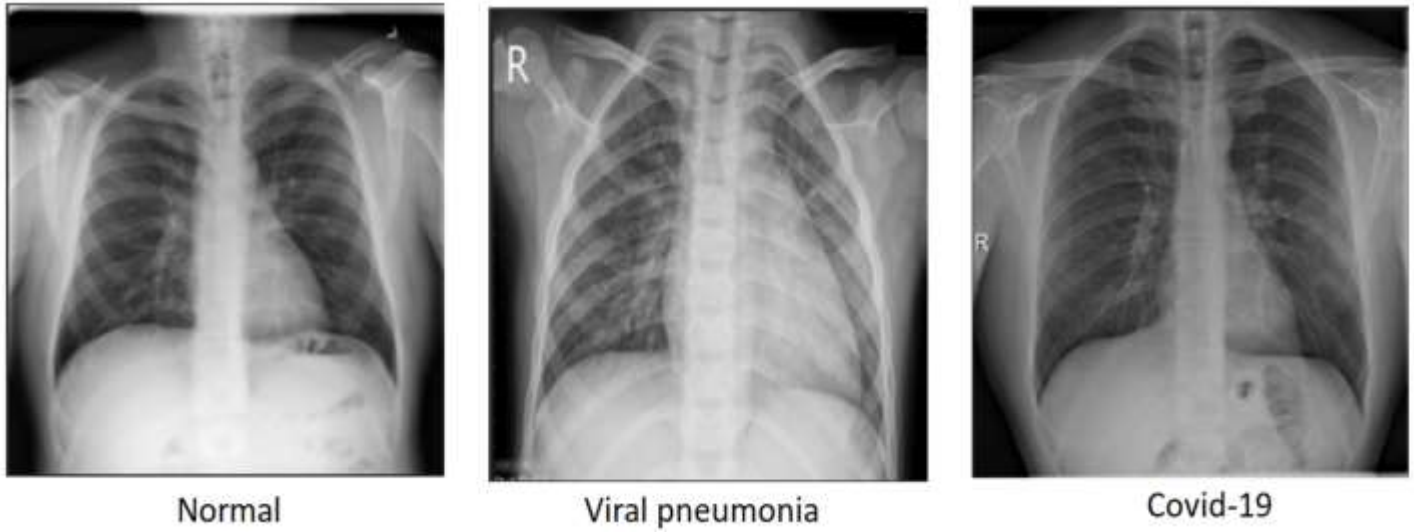
Few examples of applications where supervised learning are used:

Detecting objects in a self driving car



Source: https://www.velaninfo.com/rs/data-annotation-autonomous-vehicles-performance/

Medical image classification



Source: Farghaly, O., & Deshpande, P. (2024). Texture-Based Classification to Overcome Uncertainty between COVID-19 and Viral Pneumonia Using Machine Learning and Deep Learning Techniques. Diagnostics, 14(10), 1017. https://doi.org/10.3390/diagnostics14101017

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

Anatomy of a Supervised Learning Problem: The Dataset

UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Artificial Intelligence

What is a supersized learning dataset?

- In this section, the UCI Diabetes Dataset will be used. (https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_diabetes.html)
- Additional data columns besides bmi are age, sex, and blood pressure.
- sklearn will be used to learn more information on this dataset.

age	age in years
sex	
bmi	body mass index
bp	average blood pressure
s1	tc, T-Cells (a type of white blood cells)
s2	ldl, low-density lipoproteins
s3	hdl, high-density lipoproteins
s4	tch, thyroid stimulating hormone
s5	ltg, lamotrigine
s6	glu, blood sugar level

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

A Supervised Learning Dataset: Notation

- Training dataset of size n (e.g., n patients) is a set:

$$D = \left\{\left(x^{(i)}, y^{(i)}\right) \middle| \, i = 1, 2, \ldots, n\right\}$$

- Each $x^{(i)}$ denotes an input (e.g., the measurements for patient i), and each $y^{(i)} \in \mathcal{Y}$ is a target (e.g., the diabetes risk).

- Together, $\left(x^{(i)}, y^{(i)}\right)$ form a training example.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Training Dataset: Inputs

- An input $x^{(i)} \in \mathcal{X}$ is a *d*-dimensional vector of the form:

$$x^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_d^{(i)} \end{bmatrix}$$

- $x^{(i)}$ could be vector of attributes describing patient i. One of these attributes could be their BMI.

- The set $\mathcal{X}$ is called the feature space. Often, we have, $\mathcal{X} = \mathbb{R}^d$.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Training Dataset: Attributes

The numerical variables that describe the patient are known as attributes.

- The age of a patient.
- The patient's gender.
- The patient's BMI.

A vector of attributes, *x* could be comprised of:-

- All the pixel of an image.
- Indicator variables for whether certain words are present or absent in an input sentence.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Training Dataset: Features

An input object may have many attributes. These attributes are used to define more complex characteristics of the input. For example:

- Is the patient old and a man?
- Is the BMI above the obesity threshold?

These custom attributes are called features.

The features may be denoted via a function that takes an input and outputs a ***p***-dimensional vector:

Featurized input

$$\emptyset\left(x^{(i)}\right) = \begin{bmatrix} \emptyset\left(x^{(i)}\right)_1 \\ \emptyset\left(x^{(i)}\right)_2 \\ \vdots \\ \emptyset\left(x^{(i)}\right)_p \end{bmatrix}$$

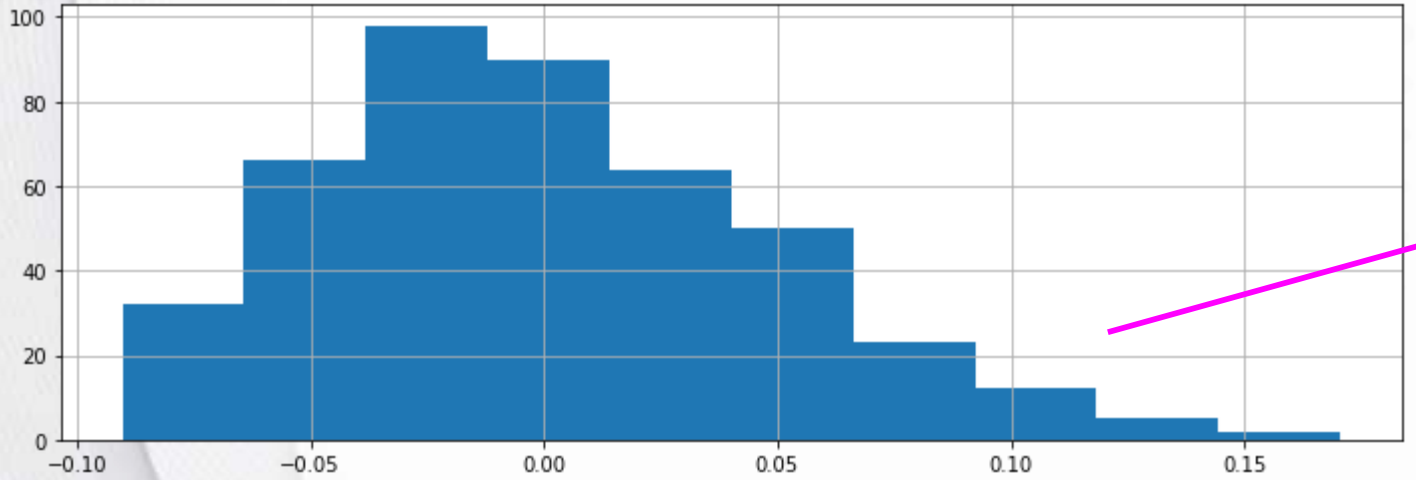
Feature

The terms attribute and features are often used interchangeably.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

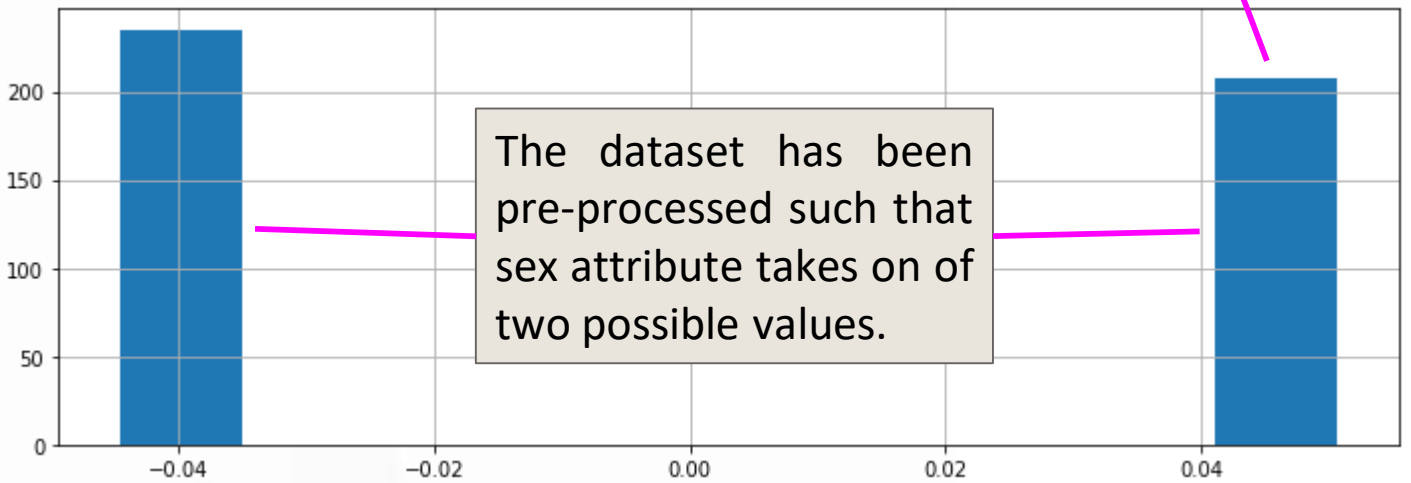
Features: Discrete vs. Continuous

- Features can be discrete or continuous.



The BMI feature is continuous.

The sex feature is discrete.



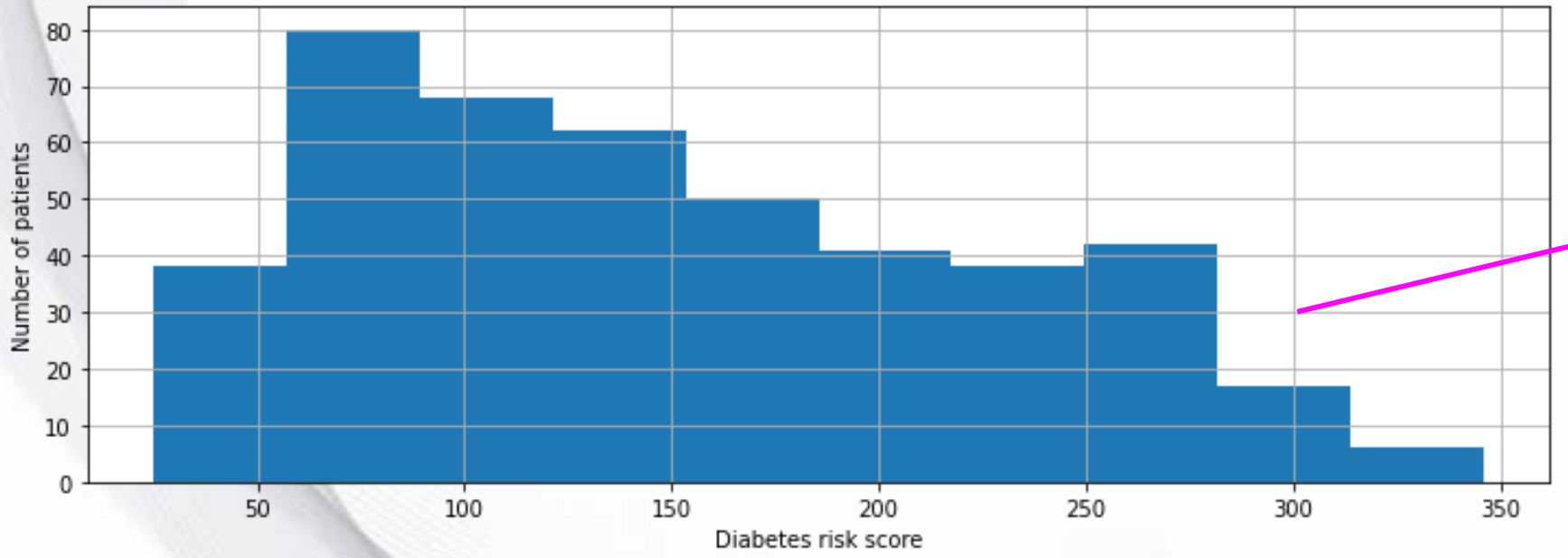
The dataset has been pre-processed such that sex attribute takes on of two possible values.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Targets

- The quantity of interest, known as the target can be predicted for each patient. In this case, this is the patient's diabetes risk.

- When $(x^{(i)}, y^{(i)})$ form a training example, each $y^{(i)} \in \mathcal{Y}$ is a target. We call $\mathcal{Y}$ the target space.



Distribution of risk scores

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Targets: Regression vs. Classification

Two types of supervised learning problems, regression and classification can be distinguished based on the form of the target variable:

- **Regression**: The target variable is continuous. We are fitting a curve in a high-dimensional feature space that approximates the shape of the dataset.

- **Classification:** The target variable is discrete. Each discrete value corresponds to a class and we are looking for a hyperplane that separates the different classes.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Feature Matrix

- The entire dataset can be represented as one matrix $\mathcal{X} \in \mathbb{R}^{n\,x\,d}$, of the form:

$$x = \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \dots & x_1^{(n)} \\ x_2^{(1)} & x_2^{(2)} & \dots & x_2^{(n)} \\ \vdots & & & \\ x_d^{(1)} & x_d^{(2)} & \dots & x_d^{(n)} \end{bmatrix}$$

- Similarly, the target variables can be vectorized into a vector $y \in \mathbb{R}^n$ of the form:

$$y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}$$

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Topic 4: Supervised Learning

- Introduction - What is supervised learning?
- Elements of a Supervised Machine Learning Problem
- A Supervised Learning Dataset
- A Supervised Learning Algorithm
- A Supervised Learning Model
- Making New Predictions
- Why Supervised Learning?
- Anatomy of a Supervised Learning Problem: The Dataset
- Anatomy of a Supervised Learning Problem: The Learning Algorithm

www.utm.my

Anatomy of a Supervised Learning Problem: The Learning Algorithm

The high-level structure of a supervised learning algorithm consisted of three components:

- A model class is the set of possible models that we consider.
- An objective function defines how good a model is.
- An optimizer finds the best predictive model in the model class according to the objective function.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Model Class

Defining a Model

A model is a function:

$$f: \mathcal{X} \to \mathcal{Y}$$

that maps inputs $x \in \mathcal{X}$ to targets $y \in \mathcal{Y}$. Often, models have parameters $\theta \in \Theta$ living in a set Θ. We will then write the model as:

$$f_\theta : \mathcal{X} \to \mathcal{Y}$$

to denote that it is parametrized by θ.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Model Class

Defining a model class

Formally, the model class is a set

$$\mathcal{M} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$$

of possible models that map input features to targets. When the models f_θ are parameterized by parameters $\theta \in \Theta$ living in some set Θ. Thus, we can also write:

$$\mathcal{M} = \{f_\theta \mid \theta \in \Theta\}$$

One simple approach is to assume that x and y are related by a linear model of the form:

$$y = \theta_\theta + \theta_1 \cdot x_1 + \theta_2 \cdot x_2 + \cdots + \theta_d \cdot x_d$$

where x is a featurized input and y is the target. The θ_j are the parameters of the model, $\Theta = \mathbb{R}^{d+1}$,and

$$\mathcal{M} = \left\{\theta_\theta + \theta_1 \cdot x_1 + \theta_2 \cdot x_2 + \cdots + \theta_d \cdot x_d \,\middle|\, \theta \in \mathbb{R}^{d+1}\right\}$$

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Objective

Given a training set, how do we pick the parameters θ for the model? A natural approach is to select θ such that $f_\theta\left(x^{(i)}\right)$ is close to $y^{(i)}$ on a training dataset $\mathcal{D} = \left\{\left(x^{(i)}, y^{(i)}\right) \middle| \, i = 1, 2, \ldots, n\right\}$.

Notation

To capture this intuition, we define an objective function (also called a *loss function*)

$$J(f) : \mathcal{M} \rightarrow [0, \infty),$$

which describes the extent to which f "fits" the data $\mathcal{D} = \left\{\left(x^{(i)}, y^{(i)}\right) \middle| \, i = 1, 2, \ldots, n\right\}$.

When f is parametrized by $\theta \in \Theta$, the objective becomes a function $J(\theta) : \Theta \rightarrow [0, \infty)$.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Objective

A few examples of objective functions:

- Mean squared error:

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(f_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

- Absolute (L1) error:

$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left|f_\theta\left(x^{(i)}\right) - y^{(i)}\right|$$

These are defined for a dataset $D = \left\{\left(x^{(i)}, y^{(i)}\right) \middle| \, i = 1, 2, \ldots, n\right\}$.

Both examples measure the difference between model predictions $f_\theta\left(x^{(i)}\right)$ and the correct targets $y^{(i)}$. By minimizing these objectives, we are trying to find θ such that the predictions $\theta^\top x^{(i)}$ are as close as possible to the targets $y^{(i)}$.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Optimizer

In order to find a good model that fits the data, we also need a procedure that will compute the parameters θ that minimize our learning objective (i.e., out training loss). This is the goal of the optimizer.

An optimizer takes an objective J and a model class M and finds a model $f \in M$ with the smallest value of the objective J.

$$\min_{f \in \mathcal{M}} J(f)$$

When f is parameterized by $\theta \in \Theta$, the optimizer minimizes a function $J(\theta)$ over all $\theta \in \Theta$.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

The Optimizer

- The sklearn.linear_models.LinearRegression algorithm optimizes the MSE loss.

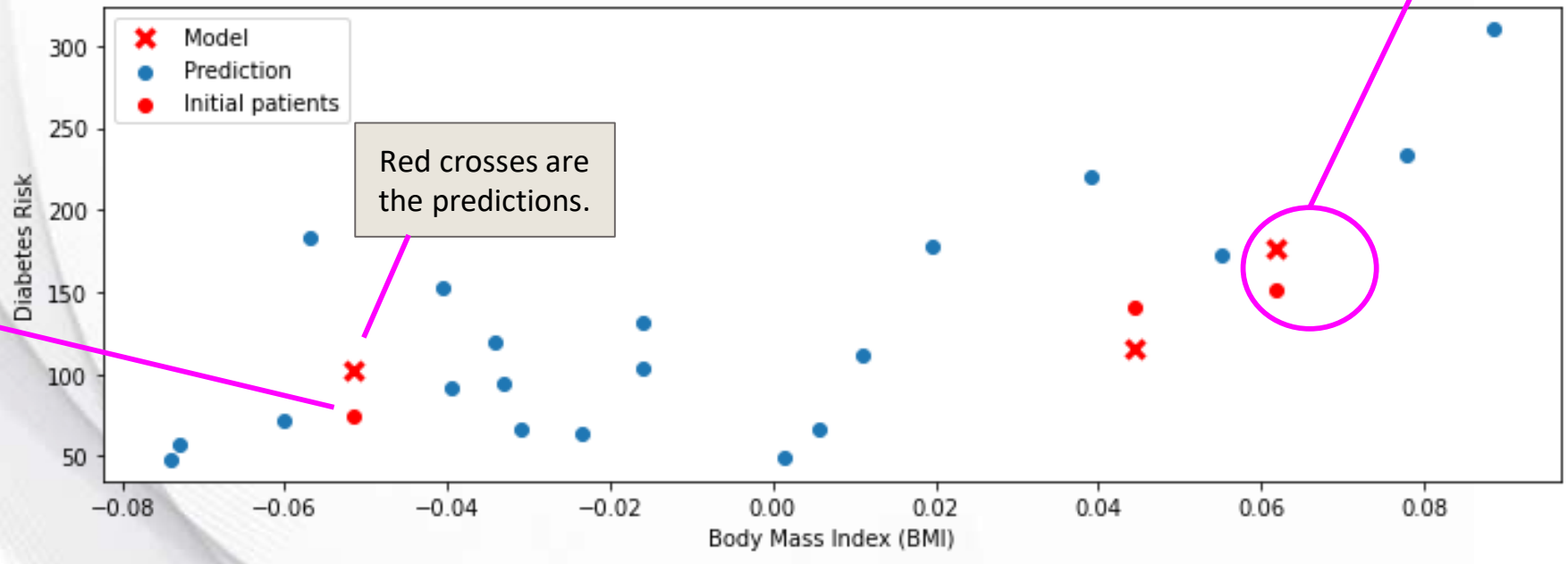
$$\min_{\theta \in \mathbb{R}} \frac{1}{2n} \sum_{i=1}^{n} \left(f_\theta\left(x^{(i)}\right) - y^{(i)} \right)^2$$

- We can measure the quality of the fit on the training set and test set.

Many additional features were used to make predictions. As a result, the predictions are more accurate. (The crosses are closer to the dots).

Red dots are the true values of y.

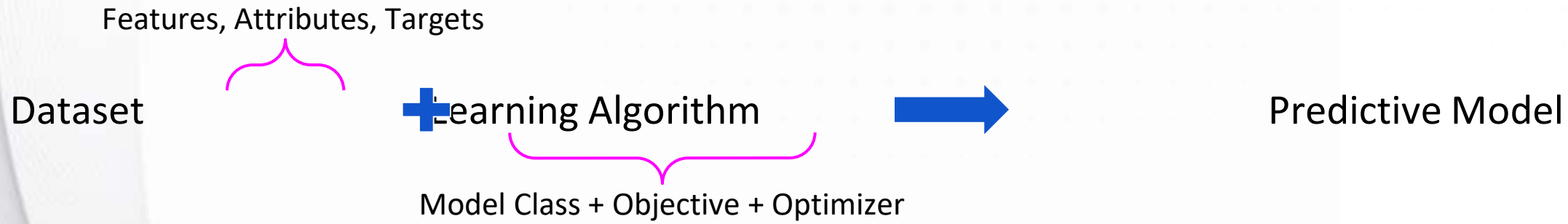
Red crosses are the predictions.



Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

Summary: Components of a Supervised Machine Learning Problem

- Formally, to apply supervised learning, we define a dataset and a learning algorithm.



- A dataset consists of training examples, which are pairs of inputs and targets. Each input is a vector of features or attributes. A learning algorithm can be fully defined by a model class, objective and optimizer.

- The output of a supervised learning is a predictive model that maps inputs to targets. For instance, it can predict targets on new inputs.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture1-introduction.html#supervised-learning

THANK YOU

univteknologimalaysia utm.my utmofficial